GLASGOW COLLEGE UESTC

Exam paper

Fundamental of Analogue Circuits (UESTC2005)

Date: (remember to complete when info available from Ruoli)
Time: (remember to complete when info available from Ruoli)

Attempt all PARTS. Total 100 marks

Use one answer sheet for each of the questions in this exam.
Show all work on the answer sheet.

Make sure that your University of Glasgow and UESTC Student Identification Numbers are on all answer sheets.

An electronic calculator may be used provided that it does not allow text storage or display, or graphical display.

DATA/FORMULAE SHEET IS PROVIDED AT THE END OF PAPER

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Q1 Consider the following circuit, where *vi* is a small-amplitude input signal, R is a 1kΩ resistor, C is a capacitor with very large capacitance, V is a 5V DC source, and D is a diode with forward voltage drop of 0.7V when conducting.

(a) Select an appropriate diode model to calculate the DC current I_D through the diode D and the DC output voltage V_D. [6]

(b) Sketch the small-signal equivalent circuit of the circuit. Derive the small signal resistance r_d and small signal gain v_d/v_i. [7]

(c) Using the superposition theorem, determine and sketch the instantaneous output voltage waveform v_D on the diode if the input is $v_i(t) = 0.01\sin(2\pi\times10^5 t)$. [8]

(d) If C is not ideal (finite capacitance), determine whether the circuit exhibits low-pass, high-pass, or band-pass characteristics for voltage on diode. Justify your reasoning. [4]

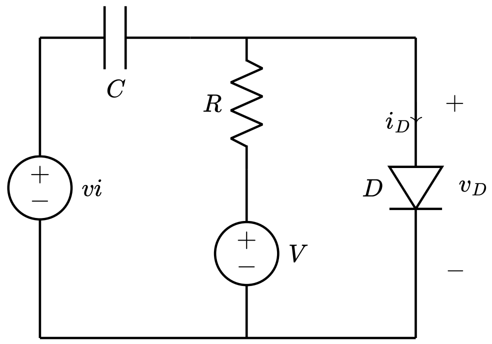

Figure Q1.

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Q2 A voltage amplifier shown as Figure Q2 can output the amplified signal to the 4kΩ load resistor R_L using one N-channel MOSFET with parameters k_n=1mA/V², Vt = 1V, λ = 0, a 5 V DC power supply V_{DD}. R=1MΩ.

(a) Specify the value of R_s so that the bias points I_D=0.5mA. [6]

(b) Choose one of the circuit's MOSFET configuration (common source, common gate, or common drain) and sketch the input signal in Figure Q2. Explain your choice with detailed reasoning. [6]

(c) Perform small-signal analysis to determine the small signal gain. [7]

(d) Calculate the input and output resistances of circuit. Evaluate the advantages and disadvantages of the amplifier. [6]

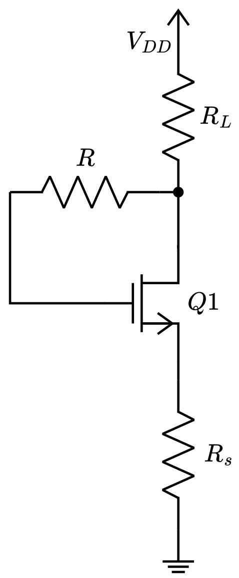

Figure Q2.

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Q3 Refer to the 2-stage amplifier circuit in Figure Q3 where the NMOSFETs are characterized by $k_n = 8\text{mA/V}^2$, $V_{tn} = 2.0\text{V}$, and the channel-length modulation can be neglected. V_{DD}=5V.

(a) Perform DC analysis to calculate each NMOSFETs' V_D, V_G, V_S, and I_D. [8]

(b) Sketch the small-signal equivalent model of the circuit. [6]

(c) Using the equivalent model, determine the overall small-signal voltage gain v_o/v_s. [6]

(d) Determine the input and output resistance of the amplifier and evaluate the total performance of the amplifier. [5]

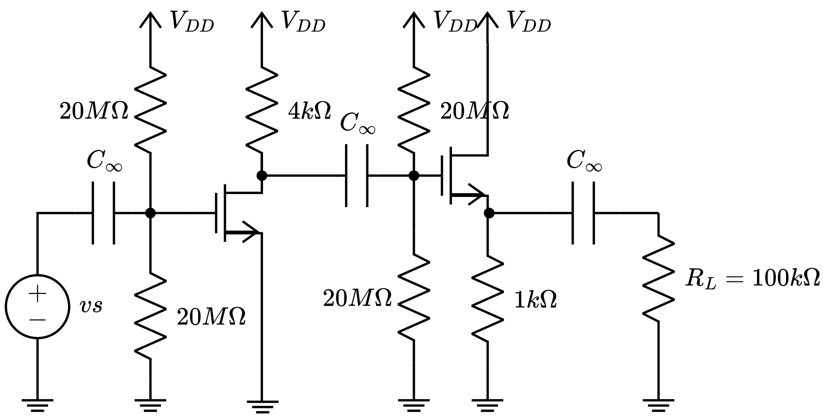

Figure Q3.

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Q4 A feedback amplifier uses ideal operational amplifier (infinite input resistance, infinite gain, zero output resistance) and a NMOSFET in saturation region as shown in Figure Q4. $R1$=5kΩ, $R2$=10kΩ, Rf=80kΩ and load resistance R_L=2kΩ.

(a) What is the topology of the feedback? Explain your reasoning. [5]

(b) Calculate the feedback factor β. [5]

(c) Determine the closed-loop gain $A_f = Vo/Vs$. [5]

(d) Find the input resistance R_{in} of the feedback amplifier. [5]

(e) Find the output resistance R_{out} of the feedback amplifier. [5]

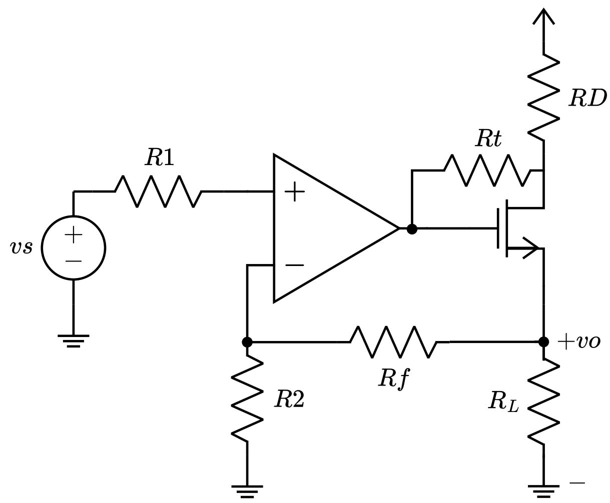

Figure Q4.

End of question paper